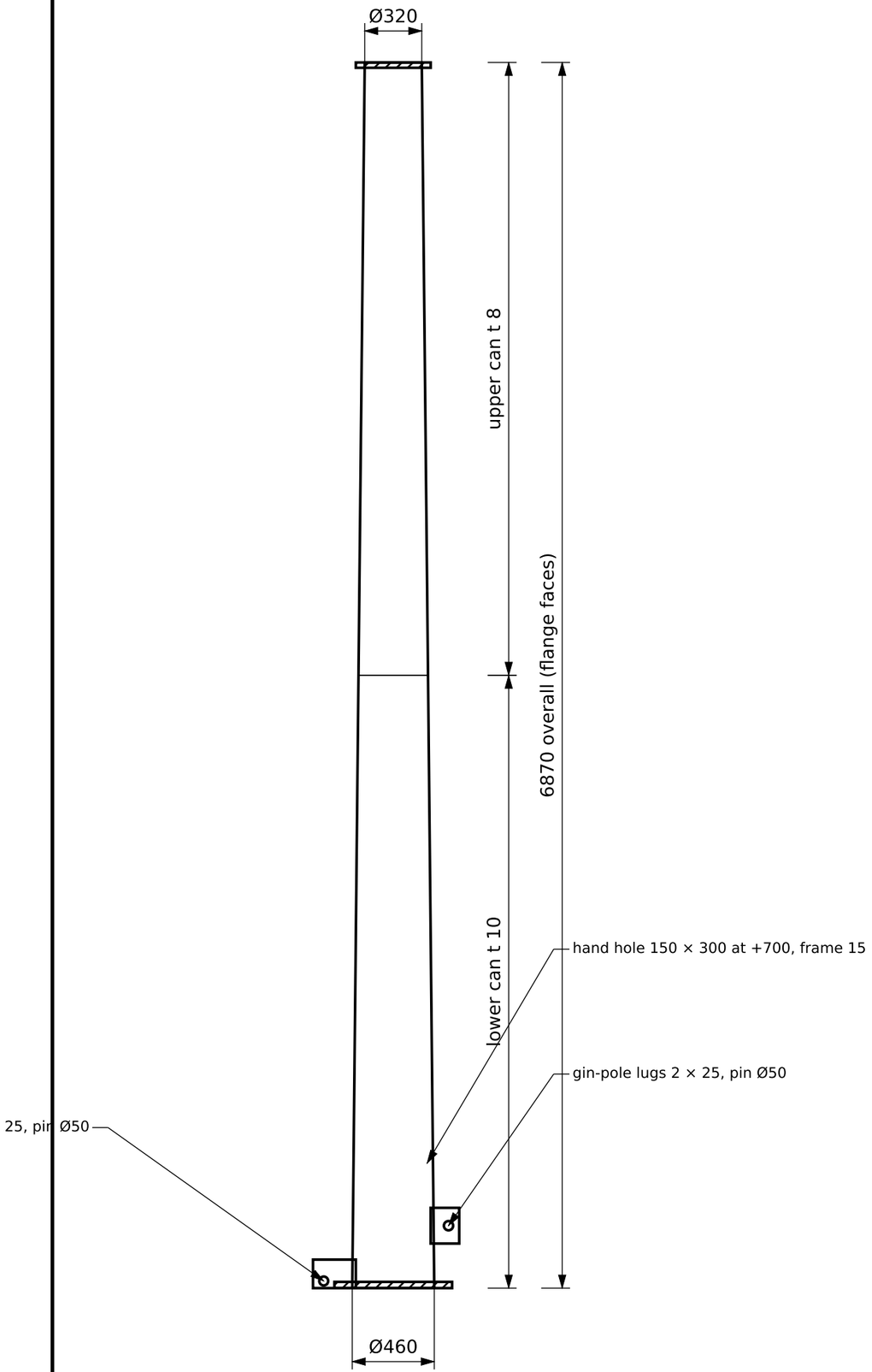
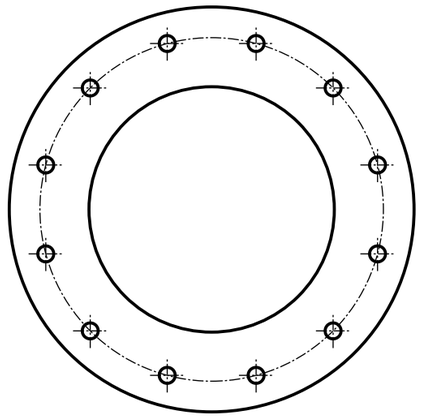
ELEVATION

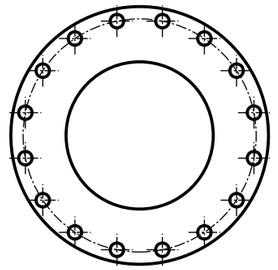


Item	Value
Shell	Ø460 × 10 → Ø320 × 8, linear taper, D/t ≤ 48.3
Cans	2 × 3.44 m rolled, one circumferential butt weld
f1 (rigid / soft+ice)	3.5 / 3.28 Hz; f2 24.04 Hz
Window	f1 ≥ 1.15 × 1P max (3.14 Hz); 3P crosses at 60–77 rpm
Governing	earthquake 0.5 g (ground D) + operation, use 0.445
Base moment (design)	238.6 kNm; SLM sensitivity use 0.601
Base bolts	12 × M24 10.9 anchors: 184.6 / 254.2 kN
Fatigue (base / mid)	0.047 / 0.0245 (turbulence × 1.234)
Tilt-up	base moment 114.7 kNm, use 0.214

FLANGES (plan)

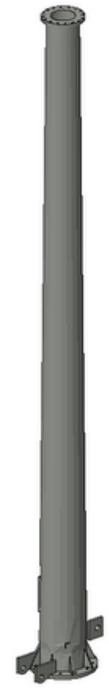


BASE FLANGE 35: 12 × Ø26 PCD 560



TOP FLANGE 30: 16 × Ø22 PCD 380 (to NY-400)

AXONOMETRIC VIEW (not to scale)



NOTES

1. Rev B replaces the 18 m two-section tower. At 8 m a soft tower (f1 below 3P at 41 rpm) is impossible with the 983 kg head, so the tower is stiff: f1 above 1P at full speed; the controller skips the 3P crossing zone like the old 1P zone.
2. Seam and can welds full penetration, ISO 5817 B, 100 % UT on flange and can welds; flanges machined flat after welding (0.2 mm), top flange square to the axis within 0.5 mm/m.
3. No climbing: service by tilting the tower down (NS6-420). Earthing bosses 2 × M12 at +300. Hand-hole cover NS6-401 with EPDM gasket and 6 × M8.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-400	B · 2026-11-06	1:25 / 1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N, rolled and seam welded; hot-dip galvanized	681 kg (CAD 698)	Claude / not checked